Status	Finished
Started	Sunday, 11 October 2026, 11:29 AM
Completed	Sunday, 11 October 2026, 11:33 AM
Duration	4 mins 27 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It has no effect on the analysis process
c. It complicates the analysis process
d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **2**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Linear Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets
b. It allows for efficient parallel processing and sorting of data
c. It avoids the need for recursion
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **4**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort does not guarantee sorted order
b. Bubble Sort is less efficient for large lists
c. Bubble Sort is difficult to implement
d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It makes it easier to select items based on their relationship to the rest of the items
c. It slows down the process
d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. order()
- c. arrange()
- d. sort()

The correct answer is: sorted()

Question **7**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. order()
- b. sort()
- c. arrange()
- d. sorted()

The correct answer is: sort()

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Comparing each element with the others
- b. Dividing the input into smaller subproblems
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Finding a specific element in a list
- b. Removing duplicates from a list
- c. Merging two datasets
- d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order

Question **10**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are discarded
- c. They are compared element by element
- d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It increases the number of comparisons needed
- b. It makes the search operation slower
- c. It has no effect on the search operation
- d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. The parts are merged without sorting
- b. Each part is searched for a specific element
- c. Each part is sorted independently
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Quick Sort
- b. Merge Sort
- c. Bubble Sort
- d. Heap Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Selection sort
- c. Merge sort
- d. Quick sort

The correct answer is:
Merge sort

Question **15**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting

b. Sorting data in a single pass

c. Sorting data sequentially

d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions